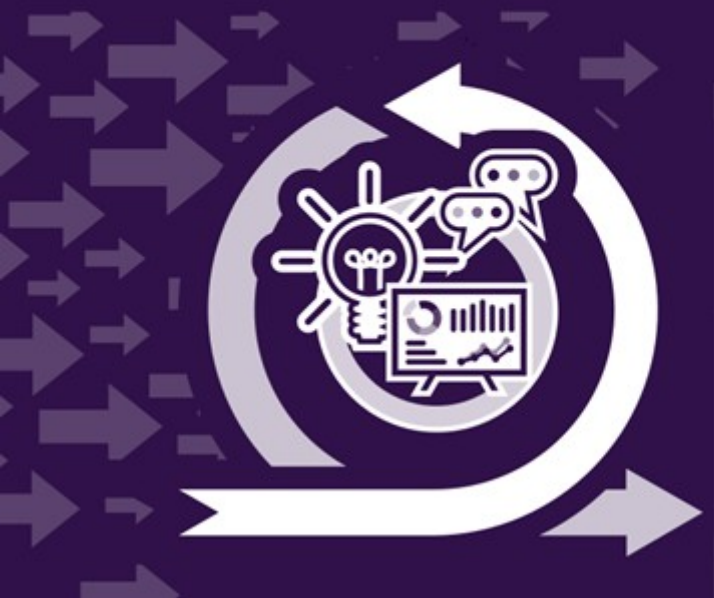
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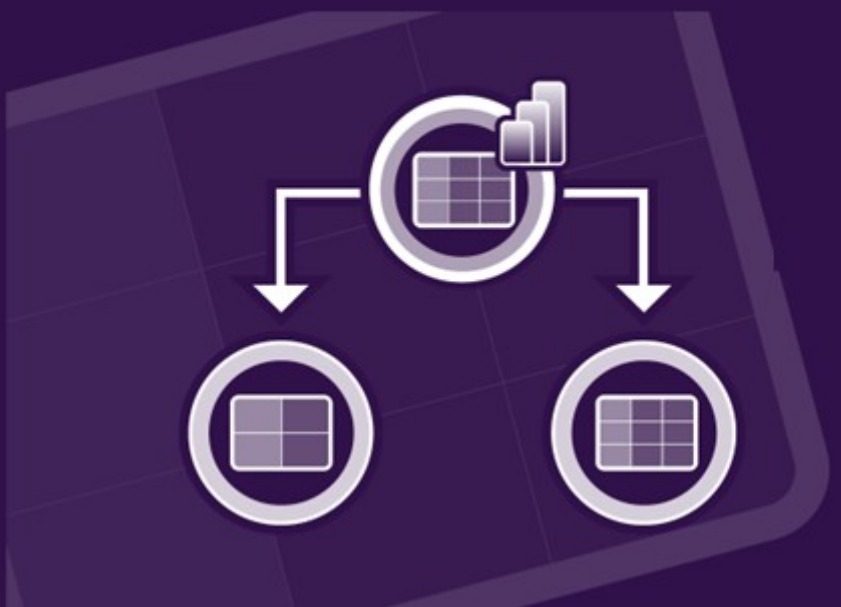
DAX filter context in Power BI

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DAX filter context in Power BI

By the end of this minibook, you'll be able to describe and work with filter context, which is used to evaluate measure formulas.

Learning objectives

By the end of this module, you'll be able to:

- Create active and inactive relationships.
- Describe filter context.
- Use the CALCULATE function to modify filter context.
- Pass filters to the CALCULATE function.
- Pass filter modifiers to the CALCULATE function.
- Perform context transition.

Exercise 01: Create relationships

Usef file 01 Start.pbix

1. In Power BI Desktop, at the left, select the **Report view** icon.



2. To view all table fields, in the **Data** pane, right-click an empty area, and then select **Expand All**.
3. To create a table visual, in the **Data** pane, from inside the **Product** table, check the **Category** field.
4. To add another column to the table, in the **Data** pane, check the **Sales | Sales** field.
5. Notice that the table visual lists four product categories, and that the sales value is the same for each, and the same for the total.

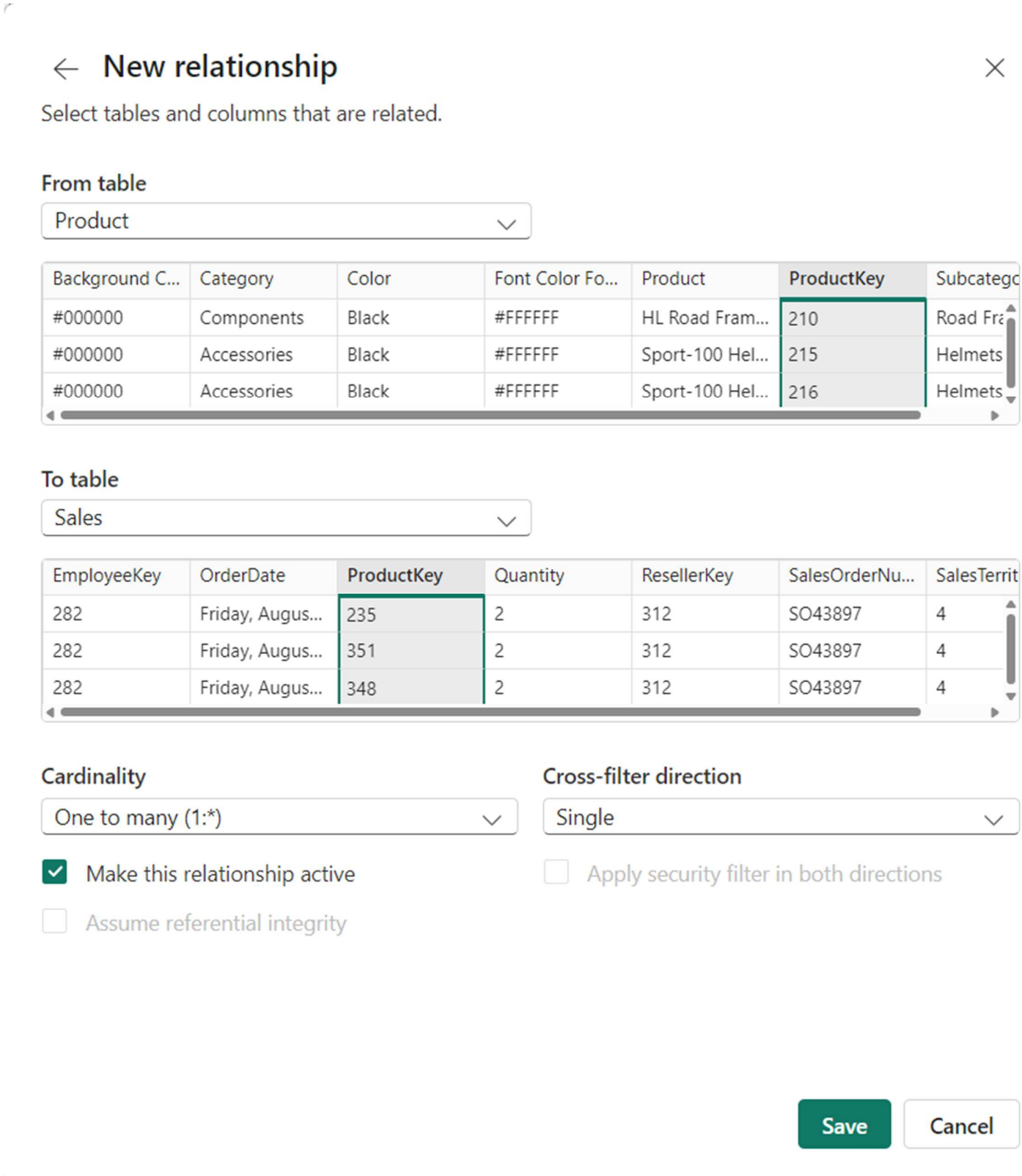
Category	Sum of Sales
Accessories	77,548,570.20
Bikes	77,548,570.20
Clothing	77,548,570.20
Components	77,548,570.20
Total	**77,548,570.20**

*The issue is that the table is based on fields from different tables. The expectation is that each product category displays the sales for that category. However, because there isn't a model relationship between these tables, the **Sales** table isn't filtered. You'll now add a relationship to propagate filters between the tables.*

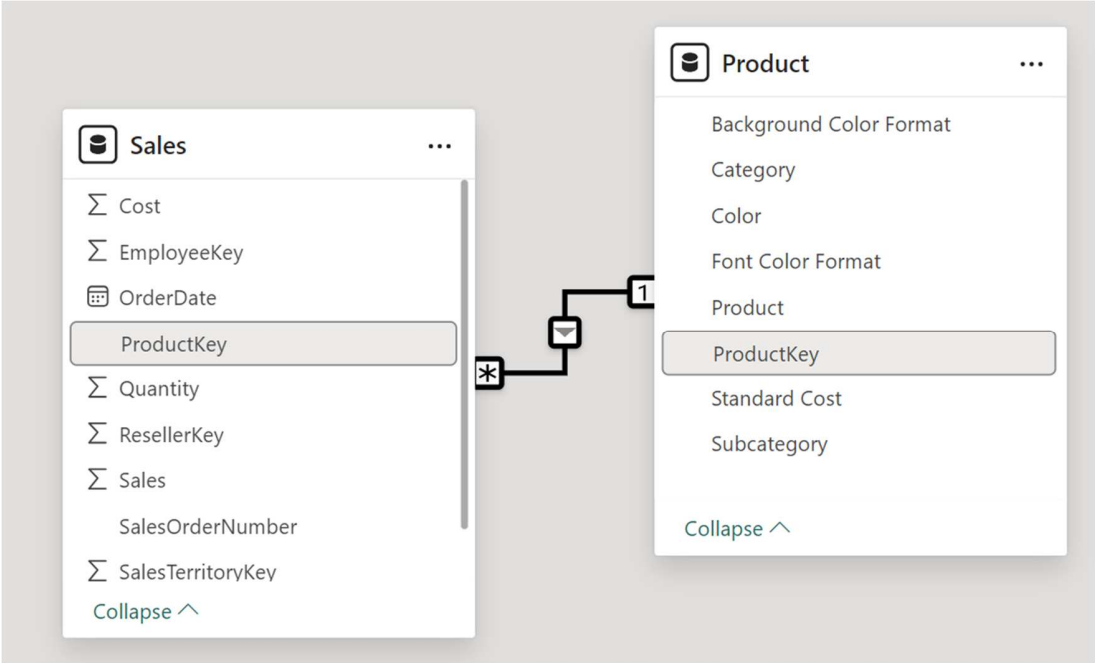
6. Select the **Model** view icon from the left navigation pane and select **Manage Relationships**.



7. In the **Manage Relationships** window, notice that no relationships are yet defined. To create a relationship, select **New relationship**.
8. Configure the relationship from **Product** table to **Sales** table. Notice the following elements were automatically configured:
    - **ProductKey columns in each table are selected**. *The columns were selected because they share the same name and data type. You may need to find matching columns with different names in real data.*
    - **Cardinality type is One To Many (1:*)**. *The cardinality was automatically detected, because Power BI understands that the **ProductKey** column from the **Product** table contains unique values. One-to-many relationships are the most common cardinality, and all relationship you create in this lab will be this type.*
    - **Cross Filter Direction type is Single**. *Single filter direction means that filters propagate from the "one side" to the "many side". In this case, it means filters applied to the **Product** table will propagate to the **Sales** table, but not in the opposite direction.*
    - **Make This Relationship Active is checked**. *Active relationships propagate filters. It's possible to mark a relationship as inactive so filters don't propagate. Inactive relationships can exist when there are multiple relationship paths between tables. In this case, model calculations can use special functions to activate them.*


New relationship
Select tables and columns that are related.
From table
Product
Background C...
Category
Color
Font Color Fo...
Product
ProductKey
Subcategc
#000000
Components
Black
#FFFFFF
HL Road Fram...
210
Road Fra
#000000
Accessories
Black
#FFFFFF
Sport-100 Hel...
215
Helmets
#000000
Accessories
Black
#FFFFFF
Sport-100 Hel...
216
Helmets
To table
Sales
EmployeeKey
OrderDate
ProductKey
Quantity
ResellerKey
SalesOrderNu...
SalesTerrit
282
Friday, Augus...
235
2
312
SO43897
4
282
Friday, Augus...
351
2
312
SO43897
4
282
Friday, Augus...
348
2
312
SO43897
4
Cardinality
One to many (1:*)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
Save
Cancel

9. Select **OK**, notice in the **Manage Relationships** window that the new relationship is listed, and then select **Close**.



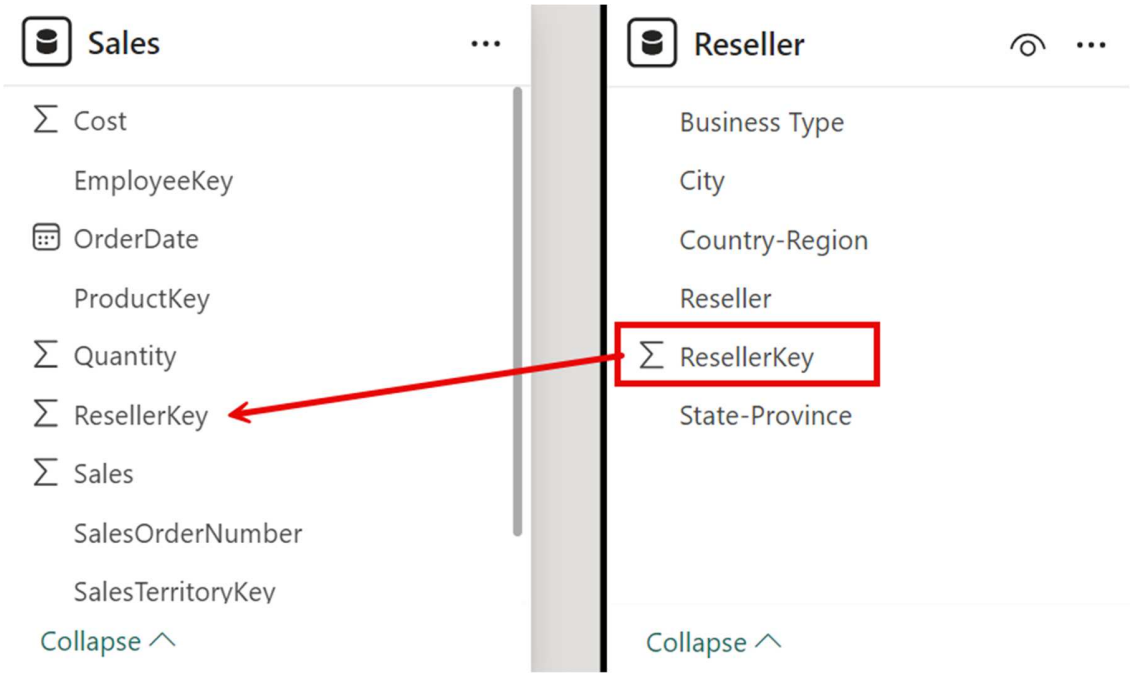
Notice there's now a connector between the two tables (*it doesn't matter if the tables are positioned next to each other*). - You can interpret the cardinality that is represented by the **1** and **(*)** indicators. - Filter direction is represented by the arrow head. - A solid line represents an active relationship; a dashed line represents an inactive relationship. - Hover the cursor over the relationship to highlight the related columns.

Create additional relationships

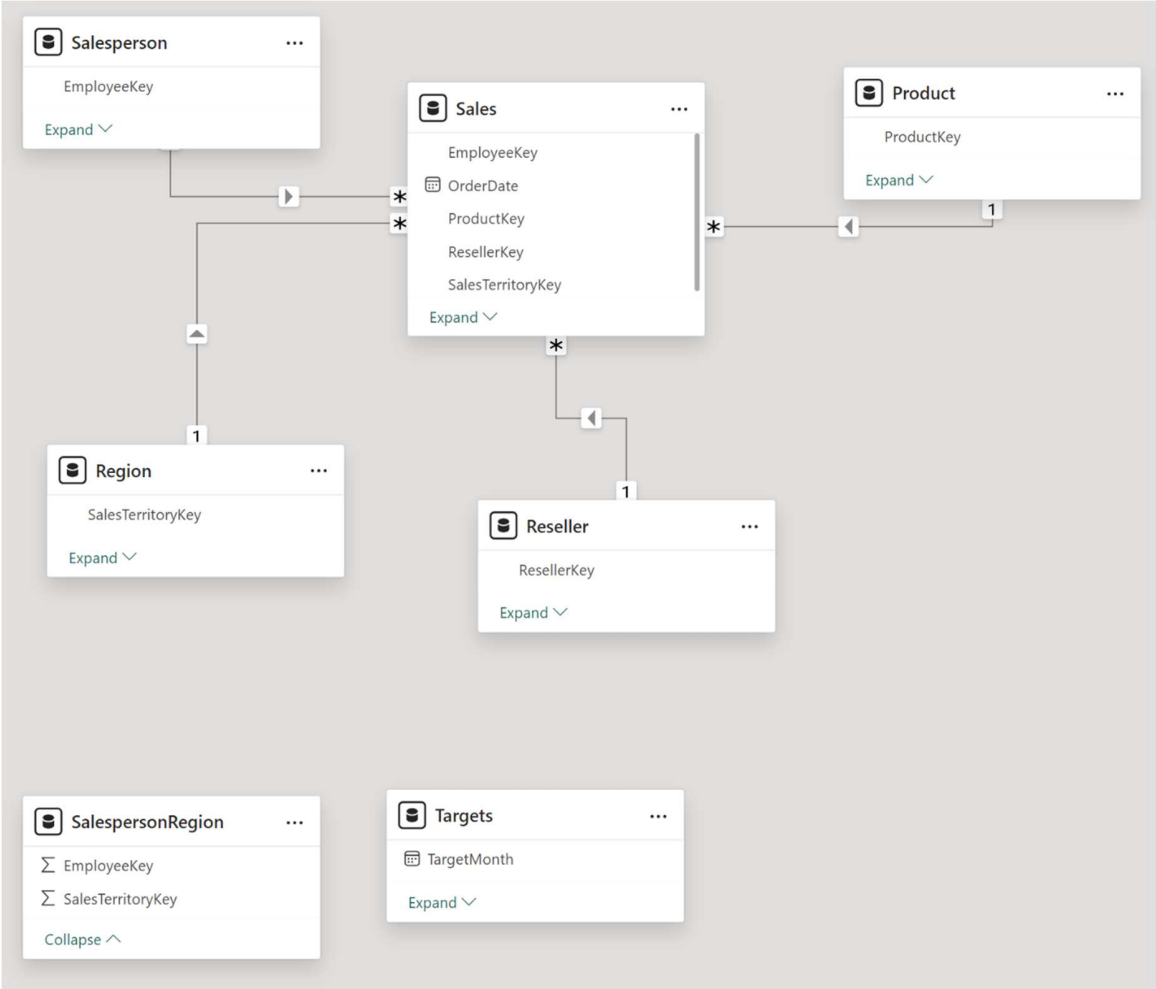
There’s an easier way to create a relationship. In the model diagram, you can drag and drop columns to create a new relationship.

1. To create a new relationship using a different technique, from the **Reseller** table, drag the **ResellerKey** column onto the **ResellerKey** column of the **Sales** table.

> ***Tip**: Sometime a column doesn’t want to be dragged. If this situation arises, select a different column, and then select the column you intend to drag again, and then try again. Ensure that you see the new relationship added to the diagram.*



2. Use the new technique to create the following two model relationships:
    - **Region | SalesTerritoryKey** to **Sales | SalesTerritoryKey**
    - **Salesperson | EmployeeKey** to **Sales | EmployeeKey**
3. In the diagram, arrange the tables so that the **Sales** table is positioned in the center of the diagram, and the related tables are arranged about it. Position the disconnected tables to the side.



4. In the report view, notice that the table visual updated to display different values for each product category.

*Filters applied to the **Product** table now propagate to the **Sales** table.*

Category	Sum of Sales
Accessories	539,106.09
Bikes	64,069,033.16
Clothing	1,714,056.05
Components	11,226,374.90
Total	**77,548,570.20**

5. **Save** the Power BI Desktop file.

Create a many-to-many relationship

In this task, you'll create a many-to-many relationship between the **Salesperson** table and the **Sales** table.

1. In Power BI Desktop, in Report view, in the **Data** pane, check the follow two fields to create a new **table** visual.
    - Salesperson | Salesperson
    - Sales | Sales

Salesperson	Sales
Amy Alberts	$737,568
David Campbell	$3,614,761
Garrett Vargas	$3,486,102
Jae Pak	$8,099,817
Jillian Carson	$9,755,992
José Saraiva	$5,536,439
Linda Mitchell	$10,158,635
Lynn Tsoflias	$1,216,917
Michael Blythe	$8,952,751
Pamela Ansman-Wolfe	$3,187,720
Rachel Valdez	$1,681,208
Total	**$77,548,570**

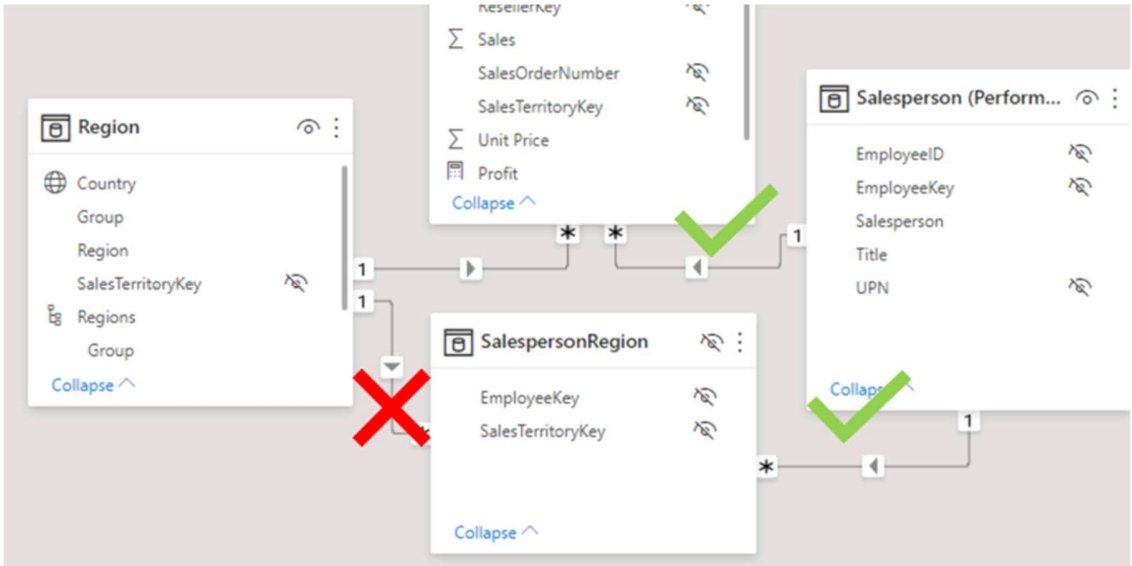
2. *The table displays sales made by each salesperson. However, there's another relationship between salespeople and sales. Some salespeople belong to one, two, or possibly more sales regions. In addition, sales regions can have multiple salespeople assigned to them.*
3. *From a performance management perspective, a salesperson's sales (based on their assigned territories) need to be analyzed and compared with sales targets. You'll create relationships to support this analysis in the next exercise.*

4. Notice that **Michael Blythe** has sold almost $9 million.
5. Switch to Model view, then drag the **SalespersonRegion** table to position it between the **Region** and **Salesperson** tables.
6. Use the drag-and-drop technique to create the following two model relationships:
    - **Salesperson | EmployeeKey** to **SalespersonRegion | EmployeeKey**
    - **Region | SalesTerritoryKey** to **SalespersonRegion | SalesTerritoryKey**

*The **SalespersonRegion** table can be considered to be a bridging table.*

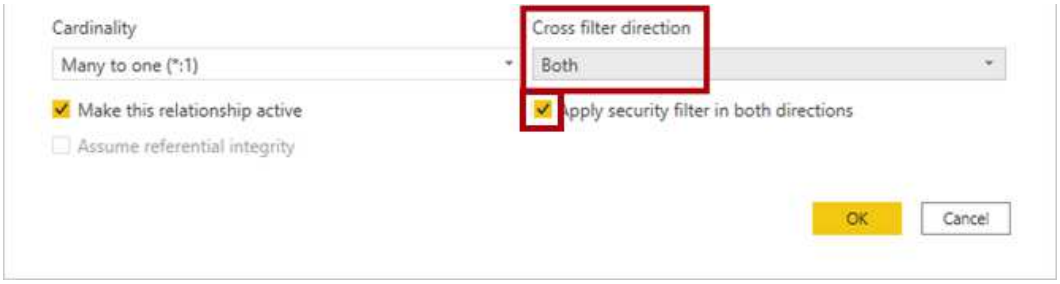
7. Switch to Report view, and then notice that the visual hasn't updated—the sales result for Michael Blythe hasn't changed.
8. Switch back to Model view, and then follow the relationship filter directions (arrowhead) from the **Salesperson** table.

*Consider that the **Salesperson** table filters the **Sales** table. It also filters the **SalespersonRegion** table, but it doesn't continue by propagating filters to the **Region** table (the arrowhead is pointing the wrong direction).*

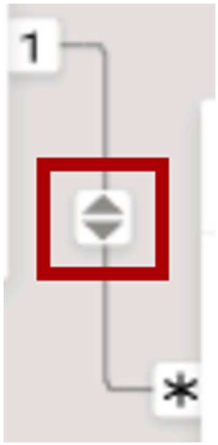


9. To edit the relationship between the **Region** and **SalespersonRegion** tables, double-click the relationship.
10. In the **Edit Relationship** window, in the **Cross Filter Direction** dropdown list, select **Both**.

11. Check the **Apply Security Filter in Both Directions** checkbox, then select OK.



12. Notice that the relationship has a double arrowhead now.



13. Switch to Report view, and then notice that the sales values have still not changed.

*The issue now relates to the fact that there are two possible filter propagation paths between the **Salesperson** and **Sales** tables. This ambiguity is internally resolved, based on a "least number of tables" assessment. To be clear, you shouldn't design models with this type of ambiguity—*

Switch to Model view to force filter propagation via the bridging table. Edit (double-click) the relationship between the **Salesperson** and **Sales** tables.

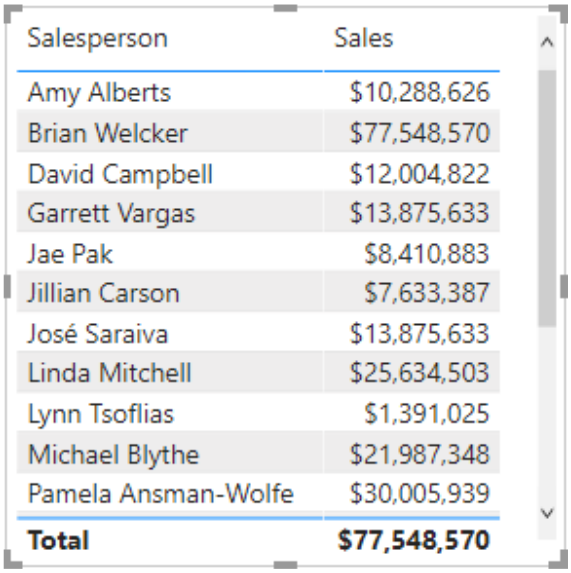
14. In the **Edit Relationship** window, uncheck the **Make This Relationship Active** checkbox, and select **OK**.

The filter propagation will now follow the only active path.

15. In the diagram, notice that the inactive relationship is represented by a dashed line.



16. Switch to Report view, and then notice that the sales for Michael Blythe are now nearly $22 million.



Salesperson	Sales
Amy Alberts	$10,288,626
Brian Welcker	$77,548,570
David Campbell	$12,004,822
Garrett Vargas	$13,875,633
Jae Pak	$8,410,883
Jillian Carson	$7,633,387
José Saraiva	$13,875,633
Linda Mitchell	$25,634,503
Lynn Tsoflias	$1,391,025
Michael Blythe	$21,987,348
Pamela Ansman-Wolfe	$30,005,939
Total	**$77,548,570**

17. Notice also, that the sales for each salesperson—if added—would exceed the table total.

> *It's a common observation of a many-to-many relationship due to the double, triple, etc. counting of regional sales results. Consider Brian Welcker, the second salesperson listed. His sales amount equals the total sales amount. It's the correct result due to the fact the he's the Director of Sales; his sales are measured by the sales of all regions.*
>
> *While the many-to-many relationship is now working, it's now not possible to analyze sales made by a salesperson (because the relationship is inactive). You'll be able to reactivate the relationship when you introduce a calculated table that will allow analyzing sales made in the sales region(s) assigned to the salesperson (for performance analysis).*

18. Switch to Model view, and then in the diagram, select the **Salesperson** table.
19. In the **Properties** pane, in the **Name** box, replace the text with **Salesperson (Performance)**.

> *The renamed table now reflects its purpose: it's used to report and analyze the performance of salespeople based on the sales of their assigned sales regions.*

Relate the Targets table

In this task, you'll create a relationship to the **Targets** table.

1. Create a relationship from the **Salesperson (Performance) | EmployeeID** column and the **Targets | EmployeeID** column.

2. In Report view, add the **Targets | Target** field to the table visual.
3. Resize the table visual so all columns are visible.



Salesperson	Sales	Target
Amy Alberts	$10,288,626	$19,450,000
Brian Welcker	$77,548,570	$221,700,000
David Campbell	$12,004,822	$19,625,000
Garrett Vargas	$13,875,633	$23,675,000
Jae Pak	$8,410,883	$13,575,000
Jillian Carson	$7,633,387	$13,675,000
José Saraiva	$13,875,633	$18,875,000
Linda Mitchell	$25,634,503	$40,850,000
Lynn Tsoflias	$1,391,025	$3,210,000
Michael Blythe	$21,987,348	$31,150,000
Pamela Ansman-Wolfe	$30,005,939	$53,850,000
Total	**$77,548,570**	**$676,210,000**

It's now possible to visualize sales and targets—but take care for two reasons. First, there's no filter on a time period, and so targets also include future target amounts. Second, targets aren't additive, and so the total shouldn't be displayed. They can either be disabled by formatting the visual or removed by using calculation logic.

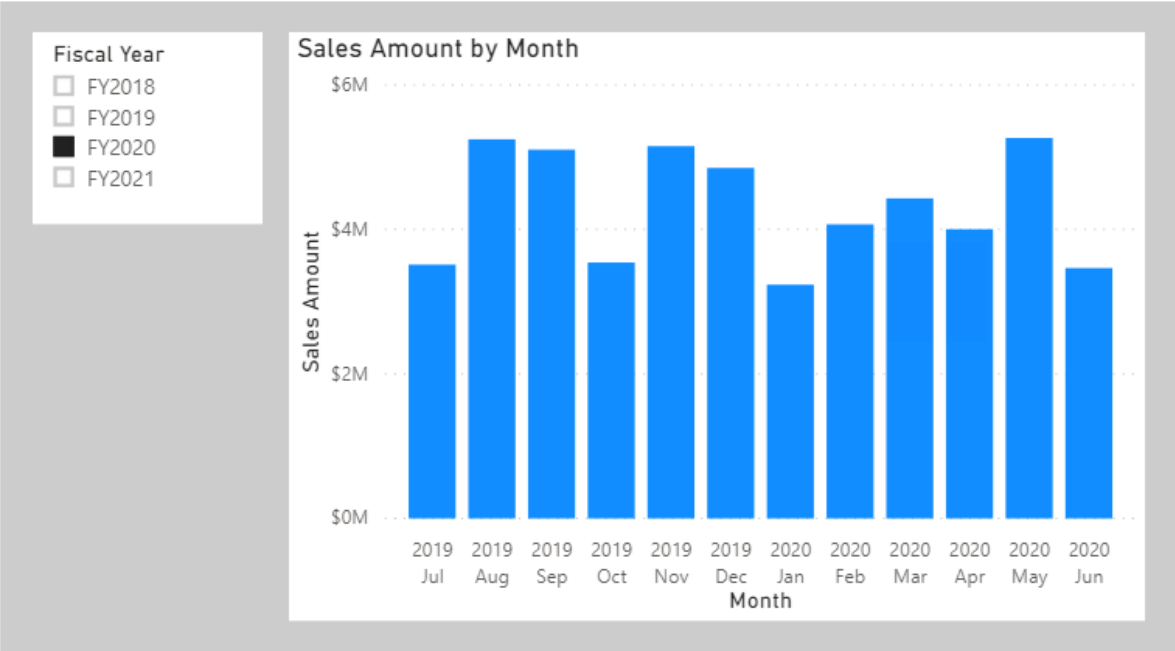
Filter Context

Filter context describes the filters that are applied during the evaluation of a measure or measure expression. Filters can be applied directly to columns, like a filter on the **Fiscal Year** column in the **Date** table for the value **FY2020**. Additionally, filters can be applied indirectly, which happens when model relationships propagate filters to other tables. For example, the **Sales** table receives a filter through its relationship with the **Date** table, filtering the **Sales** table rows to those with an **OrderDateKey** column value in **FY2020**.

Note

Calculated tables and calculated columns aren't evaluated within filter context. Calculated columns are evaluated in row context, though the formula can transition the row context to filter context, if it needs to summarize model data.

At report design time, filters are applied in the **Filters** pane or to report visuals. The slicer visual is an example of a visual whose only purpose is to filter the report page (and other pages when it's configured as a synced slicer). Report visuals, which perform grouping, also apply filters. They're implied filters; the difference is that the filter result is visible in the visual. For example, a stacked column chart visual can filter by fiscal year FY2020, group by month, and summarize sales amount. The fiscal year filter isn't visible in the visual result, yet the grouping, which results in a column for each month, behaves as a filter.



Not all filters are applied at report design time. Filters can be added when a report user interacts with the report. They can modify filter settings in the **Filters** pane, and they can cross-filter or cross-highlight visuals by selecting visual elements like columns, bars, or

pie chart segments. These interactions apply additional filters to report page visuals (unless interactions have been disabled).

It's important to understand how filter context works. It guides you in defining the correct formula for your calculations. As you write more complex formulas, you'll identify times when you need to add, modify, or remove filters to achieve the desired result.

Modify filter context

You can use the **CALCULATE** DAX function to modify filter context in your formulas. The syntax for the CALCULATE function is as follows:

```
CALCULATE(<expression>, [[<filter1>], <filter2>]…)
```

The function requires passing in an expression that returns a scalar value and as many filters as you need. The expression can be a measure (which is a named expression) or any expression that can be evaluated in filter context.

Filters can be Boolean expressions or table expressions. It's also possible to pass in filter modification functions that provide additional control when you're modifying filter context.

When you have multiple filters, they're evaluated by using the **AND** logical operator, which means that all conditions must be **TRUE** at the same time.

Problem

Consider an example that requires your formula to modify the filter context. Your objective is to produce a report visual that shows each sales region together with its revenue and revenue *as a percentage of total revenue*.

Region	Revenue	Revenue % Total Region
Australia	$10,655,335.96	9.70%
Canada	$16,355,770.46	14.89%
Central	$7,909,009.01	7.20%
France	$7,251,555.65	6.60%
Germany	$4,878,300.38	4.44%
Northeast	$6,939,374.48	6.32%
Northwest	$16,084,942.55	14.65%
Southeast	$7,879,655.07	7.18%
Southwest	$24,184,609.60	22.02%
United Kingdom	$7,670,721.04	6.99%
Total	**$109,809,274.20**	**100.00%**

The **Revenue % Total Region** result is achieved by defining a measure expression that's the ratio of revenue divided by revenue *for all regions*. Therefore, for Australia, the ratio is 10,655,335.96 dollars divided by 109,809,274.20 dollars, which is 9.7 percent.

The numerator expression doesn't need to modify filter context; it should use the current filter context (a visual that groups by region applies a filter for that region). The denominator expression, however, needs to remove any region filters to achieve the result for all regions.

Tip

The key to writing complex measures is to master these concepts:

- Understanding how filter context works.
- Understanding when and how to modify or remove filters to achieve a required result.
- Composing a formula to accurately and efficiently modify filter context.

Boolean expression filters

A Boolean expression filter is an expression that evaluates to `TRUE` or `FALSE`. Boolean filters must abide by the following rules:

- They can reference only a single column.
- They cannot reference measures.
- They cannot use functions that scan or return a table that includes aggregation functions like `SUM`.

Exercise 02: Apply Boolean expression filters

Use file 02 Start.pbix

1. Create a Measure Revenue:

```
Revenue = SUM(Sales[Sales Amount])
```

2. Create Table of Region and Revenue of each.

Region	Revenue
Australia	$10,655,335.96
Canada	$16,355,770.46
Central	$7,909,009.01
France	$7,251,555.65
Germany	$4,878,300.38
Northeast	$6,939,374.48
Northwest	$16,084,942.55
Southeast	$7,879,655.07
Southwest	$24,184,609.60
United Kingdom	$7,670,721.04
Total	**$109,809,274.20**

3.

4. Add the following measure to the **Sales** table that filters the **Revenue** measure by using a Boolean expression filter for red products.

```
Revenue Red = CALCULATE([Revenue], 'Product'[Color] = "Red")
```

5. Add the **Revenue Red** measure to the table.

Region	Revenue	Revenue Red
Australia	$10,655,335.96	$2,681,324.79
Canada	$16,355,770.46	$3,573,412.99
Central	$7,909,009.01	$1,585,997.34
France	$7,251,555.65	$1,051,014.15
Germany	$4,878,300.38	$670,607.30
Northeast	$6,939,374.48	$1,876,016.33
Northwest	$16,084,942.55	$2,292,905.61
Southeast	$7,879,655.07	$1,457,221.07
Southwest	$24,184,609.60	$5,345,637.47
United Kingdom	$7,670,721.04	$1,063,753.75
Total	**$109,809,274.20**	**$21,597,890.81**

6. Create a measure that filters the **Revenue** measure by multiple colors. Notice the use of the `IN` operator followed by a list of color values.

```
Revenue Red or Blue = CALCULATE([Revenue], 'Product'[Color]
                      IN {"Red", "Blue"})
```

7. Create a measure filters the **Revenue** measure by expensive products. Expensive products are those with a list price greater than USD 1000.

```
Revenue Expensive Products = CALCULATE([Revenue],
          'Product'[List Price] > 1000)
```

Table expression filters

A table expression filter applies a table object as a filter. It could be a reference to a model table; however, it's likely a DAX function that returns a table object.

Commonly, you'll use the **FILTER** DAX function to apply complex filter conditions, including those that can't be defined by a Boolean filter expression. The `FILTER` function is classed as an iterator function, and so you would pass in a table, or table expression, and an expression to evaluate for each row of that table.

The `FILTER` function returns a table object with exactly the same structure as one that the table passed in. Its rows are a subset of those rows that were passed in, meaning the rows where the expression evaluated as `TRUE`.

Exercise 03: Apply table expression filters

Use file 02 Start.pbix

1. Create a table filter expression that uses the `FILTER` function and add to table:

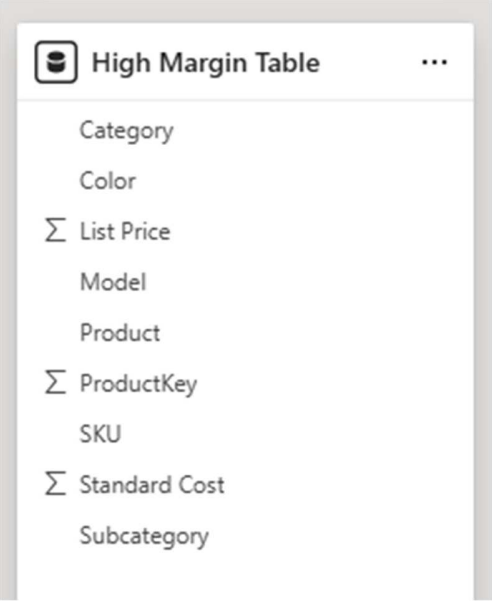
```
Revenue High Margin Products =
CALCULATE(
    [Revenue],
    FILTER(
        'Product',
        'Product'[List Price] > 'Product'[Standard Cost] * 2
    )
)
```

In this example, the `FILTER` function filters all rows of the Product table that are in filter context. Each row for a product where its list price exceeds double its standard cost is displayed as a row of the filtered table. Therefore, the **Revenue** measure is evaluated for all products that are returned by the `FILTER` function.

2. Now go to Model View and create a Calculated table with filter function to see by yourself that Filter get back table

```
High Margin Table =
    FILTER(
        'Product',
        'Product'[List Price] > 'Product'[Standard Cost] * 2
    )
```

It will return a table go to table view to see




```
High Margin Table =

    FILTER(
        'Product',
        'Product'[List Price] > 'Product'[Standard Cost] * 2
    )
```

ProductKey	Product	Standard Cost	Color	List Price	Model	Subcategory	Category	SKU
215	Sport-100 Helmet, Black	$12.0278	Black	$33.6442	Sport-100	Helmets	Accessories	HL-U509
216	Sport-100 Helmet, Black	$13.8782	Black	$33.6442	Sport-100	Helmets	Accessories	HL-U509
217	Sport-100 Helmet, Black	$13.0863	Black	$34.99	Sport-100	Helmets	Accessories	HL-U509
410	LL Mountain Front Wheel	$26.9708	Black	$60.745	LL Mountain Front Wheel	Wheels	Components	FW-M423
411	ML Mountain Front Wheel	$92.8071	Black	$209.025	ML Mountain Front Wheel	Wheels	Components	FW-M762
412	HL Mountain Front Wheel	$133.2955	Black	$300.215	HL Mountain Front Wheel	Wheels	Components	FW-M928
413	LL Road Front Wheel	$37.9909	Black	$85.565	LL Road Front Wheel	Wheels	Components	FW-R623
414	ML Road Front Wheel	$110.2829	Black	$248.385	ML Road Front Wheel	Wheels	Components	FW-R762
415	HL Road Front Wheel	$146.5466	Black	$330.06	HL Road Front Wheel	Wheels	Components	FW-R820
416	Touring Front Wheel	$96.7964	Black	$218.01	Touring Front Wheel	Wheels	Components	FW-T905
419	LL Mountain Rear Wheel	$38.9588	Black	$87.745	LL Mountain Rear Wheel	Wheels	Components	RW-M423
420	ML Mountain Rear Wheel	$104.7951	Black	$236.025	ML Mountain Rear Wheel	Wheels	Components	RW-M762
421	HL Mountain Rear Wheel	$145.2835	Black	$327.215	HL Mountain Rear Wheel	Wheels	Components	RW-M928

All filter expressions that are passed in to the **CALCULATE** function are table filter expressions. A Boolean filter expression is a shorthand notation to improve the writing and reading experience. Internally, Microsoft Power BI translates Boolean filter expressions to table filter expressions, which is how it translates your **Revenue Red** measure definition.

3. Now Recreate Revenue Red Measure with Filter function

```
Revenue Red 2 =
CALCULATE(
    [Revenue],
    FILTER(
        'Product',
        'Product'[Color] = "Red"
    )
)
```

Region	Revenue	Revenue Red	Revenue Red 2	Revenue Red or Blue	Revenue Expensive Products	Revenue High Margin Products
Australia	$10,655,335.96	$2,681,324.7861	$2,681,324.7861	$4,010,606.6904	$9,439,437.5811	$251,897.0724
Canada	$16,355,770.46	$3,573,412.985	$3,573,412.985	$4,609,513.4183	$12,105,289.9562	$799,519.5476
Central	$7,909,009.01	$1,585,997.3409	$1,585,997.3409	$1,902,957.4294	$5,962,216.6904	$269,079.3118
France	$7,251,555.65	$1,051,014.1519	$1,051,014.1519	$2,031,114.1313	$5,638,012.4086	$316,728.7318
Germany	$4,878,300.38	$670,607.3009	$670,607.3009	$1,710,778.9448	$3,976,022.3275	$168,926.1278
Northeast	$6,939,374.48	$1,876,016.3317	$1,876,016.3317	$2,221,175.8884	$4,810,209.7226	$303,475.0678
Northwest	$16,084,942.55	$2,292,905.6143	$2,292,905.6143	$3,841,905.8472	$13,186,418.4963	$583,317.5987
Southeast	$7,879,655.07	$1,457,221.0678	$1,457,221.0678	$1,860,541.064	$6,102,750.5407	$271,117.4112
Southwest	$24,184,609.60	$5,345,637.4746	$5,345,637.4746	$7,010,034.044	$18,692,962.8172	$907,214.2798
United Kingdom	$7,670,721.04	$1,063,753.7536	$1,063,753.7536	$2,002,114.3158	$6,141,817.9022	$308,130.835
Total	**$109,809,274.20**	**$21,597,890.8068**	**$21,597,890.8068**	**$31,200,741.7736**	**$86,055,138.4428**	**$4,179,405.9839**

Filter behavior

Two possible standard outcomes occur when you add filter expressions to the **CALCULATE** function:

- If the columns (or tables) aren't in filter context, then new filters will be added to the filter context to evaluate the **CALCULATE** expression.
- If the columns (or tables) are already in filter context, the existing filters will be overwritten by the new filters to evaluate the **CALCULATE** expression.

Exercise 04: Examine Filter Behavior

Use file 02 Start.pbix

The following exercise shows how adding filter expressions to the `CALCULATE` function works.

1. Create a table with Region, Revenue and Red Revenue,
   the **Revenue Red** measure was added to a table visual that groups by region and displays revenue.

Region	Revenue	Revenue Red
Australia	$10,655,335.96	$2,681,324.79
Canada	$16,355,770.46	$3,573,412.99
Central	$7,909,009.01	$1,585,997.34
France	$7,251,555.65	$1,051,014.15
Germany	$4,878,300.38	$670,607.30
Northeast	$6,939,374.48	$1,876,016.33
Northwest	$16,084,942.55	$2,292,905.61
Southeast	$7,879,655.07	$1,457,221.07
Southwest	$24,184,609.60	$5,345,637.47
United Kingdom	$7,670,721.04	$1,063,753.75
Total	**$109,809,274.20**	**$21,597,890.81**

Because no filter is applied on the **Color** column in the **Product** table, the evaluation of the measure adds a new filter to filter context. In the first row, the value of $2,681,324.79 is for red products that were sold in the Australian region.

2. Switching the first column of the table visual from **Region** to **Color** will produce a different result because the **Color** column in the **Product** table is now in filter context.

The **Revenue Red** measure formula evaluates the **Revenue** measure by adding a filter on the **Color** column (to red) in the **Product** table. Consequently, in this visual that groups by color, the measure formula overwrites the filter context with a new filter.

Color	Revenue	Revenue Red
Black	$38,236,124.06	$21,597,890.8068
Blue	$9,602,850.97	$21,597,890.8068
Grey		$21,597,890.8068
Multi	$649,030.25	$21,597,890.8068
NA	$1,099,303.91	$21,597,890.8068
Red	$21,597,890.81	$21,597,890.8068
Silver	$19,777,339.95	$21,597,890.8068
Silver/Black	$147,483.91	$21,597,890.8068
White	$29,745.13	$21,597,890.8068
Yellow	$18,669,505.22	$21,597,890.8068
Total	**$109,809,274.20**	**$21,597,890.8068**

Use filter modifier functions

When using the `CALCULATE` function, you can pass in filter modification functions, which allow you to accomplish more than adding filters alone.

Remove filters

Use the **REMOVEFILTERS** DAX function as a `CALCULATE` filter expression to remove filters from filter context. It can remove filters from one or more columns or from all columns of a single table.

Exercise 05: Remove Filters

Use file 02 Start.pbix

1. Let us solve our main problem and remove filters to get the revenue of all total without filters by region.
2. Create new page with a table of Region Revenue measure.

Region	Revenue
Southwest	$24,184,609.60
Canada	$16,355,770.46
Northwest	$16,084,942.55
Australia	$10,655,335.96
Central	$7,909,009.01
Southeast	$7,879,655.07
United Kingdom	$7,670,721.04
France	$7,251,555.65
Northeast	$6,939,374.48
Germany	$4,878,300.38
Total	**$109,809,274.20**

3. Create a measure removing all filters on table sales to get the value of total revenue

```
Total Revenul = CALCULATE([Revenue],REMOVEFILTERS('Sales'))
```

Region	Revenue	Total Revenul
Southwest	$24,184,609.60	$109,809,274.203
Canada	$16,355,770.46	$109,809,274.203
Northwest	$16,084,942.55	$109,809,274.203
Australia	$10,655,335.96	$109,809,274.203
Central	$7,909,009.01	$109,809,274.203
Southeast	$7,879,655.07	$109,809,274.203
United Kingdom	$7,670,721.04	$109,809,274.203
France	$7,251,555.65	$109,809,274.203
Northeast	$6,939,374.48	$109,809,274.203
Germany	$4,878,300.38	$109,809,274.203
Corporate HQ		$109,809,274.203
Total	**$109,809,274.20**	**$109,809,274.203**

4. Now you can another measure to get the percentage.

```
Revenue % to Company Total = DIVIDE([Revenue],[Total Revenul],0)
```

Region	Revenue	Total Revenul	Revenue % to Company Total
Southwest	$24,184,609.60	$109,809,274.203	22.02%
Canada	$16,355,770.46	$109,809,274.203	14.89%
Northwest	$16,084,942.55	$109,809,274.203	14.65%
Australia	$10,655,335.96	$109,809,274.203	9.70%
Central	$7,909,009.01	$109,809,274.203	7.20%
Southeast	$7,879,655.07	$109,809,274.203	7.18%
United Kingdom	$7,670,721.04	$109,809,274.203	6.99%
France	$7,251,555.65	$109,809,274.203	6.60%
Northeast	$6,939,374.48	$109,809,274.203	6.32%
Germany	$4,878,300.38	$109,809,274.203	4.44%
Corporate HQ		$109,809,274.203	
Total	**$109,809,274.20**	**$109,809,274.203**	**100.00%**

5. Add a new measure to the **Sales** table that evaluates the **Revenue** measure but does so by removing filters from the **Sales Territory** table. Format the measure as currency with two decimal places.

```
Revenue Total Region = CALCULATE([Revenue],
REMOVEFILTERS('Sales Territory'))
```

Now, add the **Revenue Total Region** measure to the matrix visual that is found on **Page 2** of the report. The matrix visual will group by three columns from the Sales Territory table on the rows: **Group**, **Country**, and **Region**.

Reseller Revenue

Group	Country	Region	Revenue	Revenue Total Region
Corporate HQ	Corporate HQ	Corporate HQ		$80,450,596.98
		Total		**$80,450,596.98**
	Total			**$80,450,596.98**
Europe	France	France	$4,607,537.94	$80,450,596.98
		Total	**$4,607,537.94**	**$80,450,596.98**
	Germany	Germany	$1,983,988.04	$80,450,596.98
		Total	**$1,983,988.04**	**$80,450,596.98**
	United Kingdom	United Kingdom	$4,279,008.83	$80,450,596.98
		Total	**$4,279,008.83**	**$80,450,596.98**
	Total		**$10,870,534.80**	**$80,450,596.98**
North America	Canada	Canada	$14,377,925.60	$80,450,596.98
		Total	**$14,377,925.60**	**$80,450,596.98**
	United States	Central	$7,906,008.18	$80,450,596.98
		Northeast	$6,932,842.01	$80,450,596.98
		Northwest	$12,435,076.00	$80,450,596.98
		Southeast	$7,867,416.23	$80,450,596.98
		Southwest	$18,466,458.79	$80,450,596.98
		Total	**$53,607,801.21**	**$80,450,596.98**
	Total		**$67,985,726.81**	**$80,450,596.98**
Pacific	Australia	Australia	$1,594,335.38	$80,450,596.98
		Total	**$1,594,335.38**	**$80,450,596.98**
	Total		**$1,594,335.38**	**$80,450,596.98**
Total			**$80,450,596.98**	**$80,450,596.98**

Notice that each **Revenue Total Region** value is the same. It's the value of total revenue.

6. Create now measure **`Revenue % Total Region`** `that Divide` **`Revenue`** `by` **`Revenue Total`** `Measure.`
7. `Format as % and add to the table.`

Revenue Total Region on its own isn't useful, when it's used as a denominator in a ratio, it calculates a percent of grand total. Therefore, you will now overwrite

`Revenue % Total Region` then you can remove the **Revenue Total Region** measure definition with the following definition. (This new definition changes the measure name and declares two variables. Be sure to format the measure as a percentage with two decimal places.)

```
Revenue % Total Region =
VAR CurrentRegionRevenue = [Revenue]
VAR TotalRegionRevenue =
    CALCULATE(
        [Revenue],
        REMOVEFILTERS('Sales Territory')
    )
RETURN
    DIVIDE(
        CurrentRegionRevenue,
        TotalRegionRevenue
    )
```

Verify that the matrix visual now displays the **Revenue % Total Region** values.

Reseller Revenue

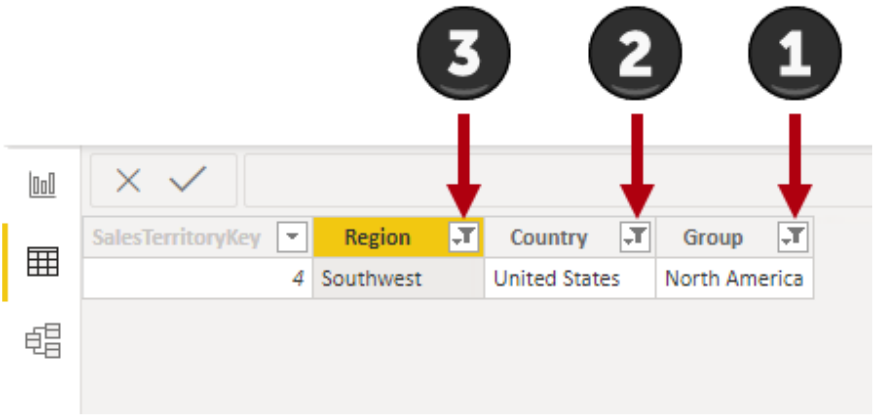
Group	Country	Region	Revenue	Revenue % Total Region
Europe	France	France	$4,607,537.94	5.73%
		Total	**$4,607,537.94**	**5.73%**
	Germany	Germany	$1,983,988.04	2.47%
		Total	**$1,983,988.04**	**2.47%**
	United Kingdom	United Kingdom	$4,279,008.83	5.32%
		Total	**$4,279,008.83**	**5.32%**
	Total		**$10,870,534.80**	**13.51%**
North America	Canada	Canada	$14,377,925.60	17.87%
		Total	**$14,377,925.60**	**17.87%**
	United States	Central	$7,906,008.18	9.83%
		Northeast	$6,932,842.01	8.62%
		Northwest	$12,435,076.00	15.46%
		Southeast	$7,867,416.23	9.78%
		Southwest	$18,466,458.79	22.95%
		Total	**$53,607,801.21**	**66.63%**
	Total		**$67,985,726.81**	**84.51%**
Pacific	Australia	Australia	$1,594,335.38	1.98%
		Total	**$1,594,335.38**	**1.98%**
	Total		**$1,594,335.38**	**1.98%**
Total			**$80,450,596.98**	**100.00%**

8. You'll now create another measure, but this time, you will calculate the ratio of revenue for a region divided by its country's or region's revenue.

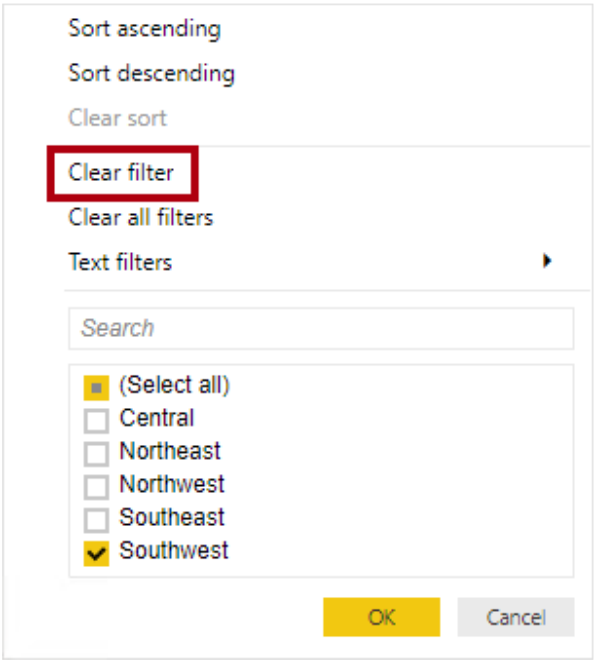
Before you complete this task, notice that the **Revenue % Total Region** value for the Southwest region is 22.95 percent. Investigate the filter context for this cell. Switch to data view and then, in the **Fields** pane, select the **Sales Territory** table.

Apply the following column filters:

- **Group** - North America
- **Country** - United States
- **Region** - Southwest



Notice that the filters reduce the table to only one row. Now, while thinking about your new objective to create a ratio of the region revenue over its country's revenue, clear the filter from the **Region** column.



Notice that five rows now exist, each row belonging to the country United States. Accordingly, when you clear the **Region** column filters, while preserving filters on the **Country** and **Group** columns, you will have a new filter context that's for the region's country.

9. In the following measure definition, notice how you can clear or remove a filter from a column. In DAX logic, it's a small and subtle change that's made to the **Revenue % Total Region** measure formula: The `REMOVEFILTERS` function

now removes filters from the **Region** column instead of all columns of the **Sales Territory** table.

```
Revenue % Total Country =
VAR CurrentRegionRevenue = [Revenue]
VAR TotalCountryRevenue =
    CALCULATE(
        [Revenue],
        REMOVEFILTERS('Sales Territory'[Region])
    )
RETURN
    DIVIDE(
        CurrentRegionRevenue,
        TotalCountryRevenue
    )
```

Add the **Revenue % Total Country** measure to the **Sales** table and then format it as a percentage with two decimal places. Add the new measure to the matrix visual.

Reseller Revenue

Group	Country	Region	Revenue	Revenue % Total Region	Revenue % Total Country
Europe	France	France	$4,607,537.94	5.73%	100.00%
		Total	**$4,607,537.94**	**5.73%**	**100.00%**
	Germany	Germany	$1,983,988.04	2.47%	100.00%
		Total	**$1,983,988.04**	**2.47%**	**100.00%**
	United Kingdom	United Kingdom	$4,279,008.83	5.32%	100.00%
		Total	**$4,279,008.83**	**5.32%**	**100.00%**
	Total		**$10,870,534.80**	**13.51%**	**100.00%**
North America	Canada	Canada	$14,377,925.60	17.87%	100.00%
		Total	**$14,377,925.60**	**17.87%**	**100.00%**
	United States	Central	$7,906,008.18	9.83%	14.75%
		Northeast	$6,932,842.01	8.62%	12.93%
		Northwest	$12,435,076.00	15.46%	23.20%
		Southeast	$7,867,416.23	9.78%	14.68%
		Southwest	$18,466,458.79	22.95%	34.45%
		Total	**$53,607,801.21**	**66.63%**	**100.00%**
	Total		**$67,985,726.81**	**84.51%**	**100.00%**
Pacific	Australia	Australia	$1,594,335.38	1.98%	100.00%
		Total	**$1,594,335.38**	**1.98%**	**100.00%**
	Total		**$1,594,335.38**	**1.98%**	**100.00%**
Total			**$80,450,596.98**	**100.00%**	**100.00%**

Notice that all values, except those values for United States regions, are 100 percent. The reason is because, at the company, the United States has regions, while all other countries/regions do not.

10. The next example is last measure that you will create. Add the **Revenue % Total Group** measure, and then format it as a percentage with two decimal places. Then, add the new measure to the matrix visual.

```
Revenue % Total Group =
VAR CurrentRegionRevenue = [Revenue]
VAR TotalGroupRevenue =
    CALCULATE(
        [Revenue],
        REMOVEFILTERS(
            'Sales Territory'[Region],
            'Sales Territory'[Country]
        )
    )
RETURN
    DIVIDE(
          CurrentRegionRevenue,
        TotalGroupRevenue
    )
```

Reseller Revenue

Group	Country	Region	Revenue	Revenue % Total Region	Revenue % Total Country	Revenue % Total Group
Europe	France	France	$4,607,537.94	5.73%	100.00%	42.39%
		Total	**$4,607,537.94**	**5.73%**	**100.00%**	**42.39%**
	Germany	Germany	$1,983,988.04	2.47%	100.00%	18.25%
		Total	**$1,983,988.04**	**2.47%**	**100.00%**	**18.25%**
	United Kingdom	United Kingdom	$4,279,008.83	5.32%	100.00%	39.36%
		Total	**$4,279,008.83**	**5.32%**	**100.00%**	**39.36%**
	Total		**$10,870,534.80**	**13.51%**	**100.00%**	**100.00%**
North America	Canada	Canada	$14,377,925.60	17.87%	100.00%	21.15%
		Total	**$14,377,925.60**	**17.87%**	**100.00%**	**21.15%**
	United States	Central	$7,906,008.18	9.83%	14.75%	11.63%
		Northeast	$6,932,842.01	8.62%	12.93%	10.20%
		Northwest	$12,435,076.00	15.46%	23.20%	18.29%
		Southeast	$7,867,416.23	9.78%	14.68%	11.57%
		Southwest	$18,466,458.79	22.95%	34.45%	27.16%
		Total	**$53,607,801.21**	**66.63%**	**100.00%**	**78.85%**
	Total		**$67,985,726.81**	**84.51%**	**100.00%**	**100.00%**
Pacific	Australia	Australia	$1,594,335.38	1.98%	100.00%	100.00%
		Total	**$1,594,335.38**	**1.98%**	**100.00%**	**100.00%**
	Total		**$1,594,335.38**	**1.98%**	**100.00%**	**100.00%**
Total			**$80,450,596.98**	**100.00%**	**100.00%**	**100.00%**

When you remove filters from the **Region** and **Country** columns in the **Sales Territory** table, the measure will calculate the region revenue as a ratio of its group's revenue.

Preserve filters

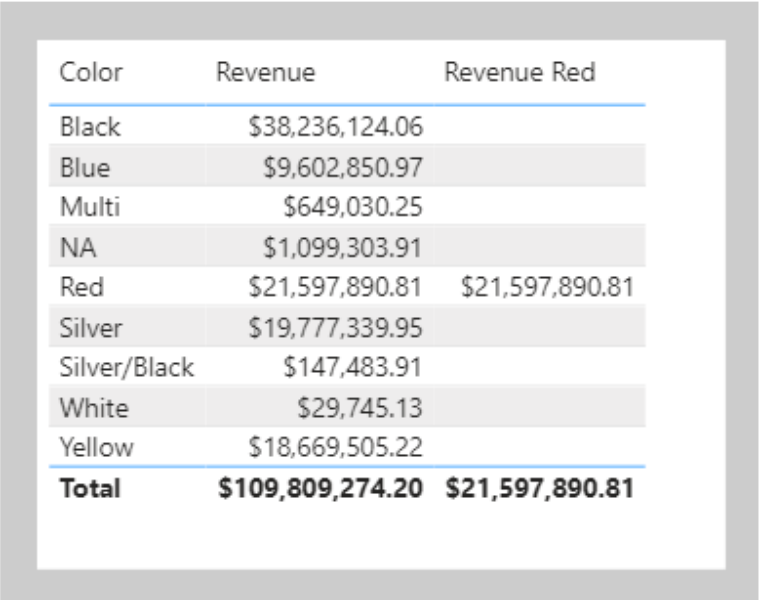
You can use the **KEEPFILTERS** DAX function as a filter expression in the `CALCULATE` function to preserve filters.

Exercise 06: Keep Filters

Use file 02 Start.pbix

To observe how to accomplish this task, switch to **Page 3** of the report. Then, modify the **Revenue Red** measure definition to use the `KEEPFILTERS` function.

```
Revenue Red =
CALCULATE(
    [Revenue],
    KEEPFILTERS('Product'[Color] = "Red")
)
```



Color	Revenue	Revenue Red
Black	$38,236,124.06	
Blue	$9,602,850.97	
Multi	$649,030.25	
NA	$1,099,303.91	
Red	$21,597,890.81	$21,597,890.81
Silver	$19,777,339.95	
Silver/Black	$147,483.91	
White	$29,745.13	
Yellow	$18,669,505.22	
Total	**$109,809,274.20**	**$21,597,890.81**

In the table visual, notice that only one **Revenue Red** value exists. The reason is because the Boolean filter expression preserves existing filters on the **Color** column in the **Product** table. The reason why colors other than red are BLANK is because the filter contexts and the filter expressions are combined for these two filters. The color black and color red are intersected, and because both can't be `TRUE` at the same time, the expression is filtered by no product rows. It's only possible that both red filters can be `TRUE` at the same time, which explains why the one **Revenue Red** value is shown.

Use inactive relationships

An inactive model relationship can only propagate filters when the **USERELATIONSHIP** DAX function is passed as a filter expression to the `CALCULATE` function. When you use this function to engage an inactive relationship, the active relationship will automatically become inactive.

Exercise 07: USERELATIONSHIP Function

Use file 02 Start.pbix

1. Create a measure definition that uses an inactive relationship to calculate the **Revenue** measure by shipped dates:

```
Revenue Shipped =
CALCULATE (
    [Revenue],
    USERELATIONSHIP('Date'[DateKey], Sales[ShipDateKey])
)
```

2. Create a table for regions , Revenue and Revenue Shipped in Page 5

Region	Revenue	Revenue Shipped
Southwest	$24,184,609.60	$24,184,609.6047
Canada	$16,355,770.46	$16,355,770.4586
Northwest	$16,084,942.55	$16,084,942.5513
Australia	$10,655,335.96	$10,655,335.9611
Central	$7,909,009.01	$7,909,009.0073
Southeast	$7,879,655.07	$7,879,655.0747
United Kingdom	$7,670,721.04	$7,670,721.0375
France	$7,251,555.65	$7,251,555.6493
Northeast	$6,939,374.48	$6,939,374.483
Germany	$4,878,300.38	$4,878,300.3755
Total	**$109,809,274.20**	**$109,809,274.203**

3. Notice there is no difference because we filter by region and Date is not included in filter.
4. Replace Regin with Fiscal Year and see the result

Year	Revenue	Revenue Shipped
FY2020	$51,878,274.54	$51,159,808.731
FY2019	$34,070,108.50	$33,886,063.4237
FY2018	$23,860,891.17	$23,482,328.6411
		$1,281,073.4072
Total	**$109,809,274.20**	**$109,809,274.203**

Examine filter context

The **VALUES** DAX function lets your formulas determine what values are in filter context.

The `VALUES` function syntax is as follows:

```
VALUES(<TableNameOrColumnName>)
```

The function requires passing in a table reference or a column reference. When you pass in a table reference, it returns a table object with the same columns that contain rows for what's in filter context. When you pass in a column reference, it returns a single-column table of unique values that are in filter context.

The function always returns a table object and it's possible for a table to contain multiple rows. Therefore, to test whether a specific value is in filter context, your formula must first test that the `VALUES` function returns a single row. Two functions can help you accomplish this task: the **HASONEVALUE** and the **SELECTEDVALUE** DAX functions.

The `HASONEVALUE` function returns `TRUE` when a given column reference has been filtered down to a single value.

The `SELECTEDVALUE` function simplifies the task of determining what a single value could be. When the function is passed a column reference, it'll return a single value, or when more than one value is in filter context, it'll return BLANK (or an alternate value that you pass to the function).

Exercise 08: HASNOVALUE Function

Use file 02 Start.pbix

In the following example, you will use the `HASONEVALUE` function.

1. In Page 6 crate a table with region and Revenue.

2. Add the following measure, which calculates sales commission, to the **Sales** table. Note that, the commission rate is 10 percent of revenue for all countries/regions except the United States.
3. In the United States, salespeople earn 15 percent commission.
4. Format the measure as currency with two decimal places, and then add it to the table that is found on **Page 3** of the report.

```
Sales Commission =
[Revenue]
    * IF(
        HASONEVALUE('Sales Territory'[Country]),
        IF(
            VALUES('Sales Territory'[Country]) = "United
States",
            0.15,
            0.1
        )
    )
```

Region	Revenue	Sales Commission
Australia	$10,655,335.96	$1,065,533.60
Canada	$16,355,770.46	$1,635,577.05
Central	$7,909,009.01	$1,186,351.35
France	$7,251,555.65	$725,155.56
Germany	$4,878,300.38	$487,830.04
Northeast	$6,939,374.48	$1,040,906.17
Northwest	$16,084,942.55	$2,412,741.38
Southeast	$7,879,655.07	$1,181,948.26
Southwest	$24,184,609.60	$3,627,691.44
United Kingdom	$7,670,721.04	$767,072.10
Total	**$109,809,274.20**	

5. Add the measure to the table.

Notice that the total **Sales Commission** result is **BLANK**. The reason is because multiple values are in filter context for the **Country** column in the **Sales Territory** table. In this

case, the **HASONEVALUE** function returns **FALSE**, which results in the **Revenue** measure being multiplied by BLANK (a value multiplied by BLANK is BLANK).

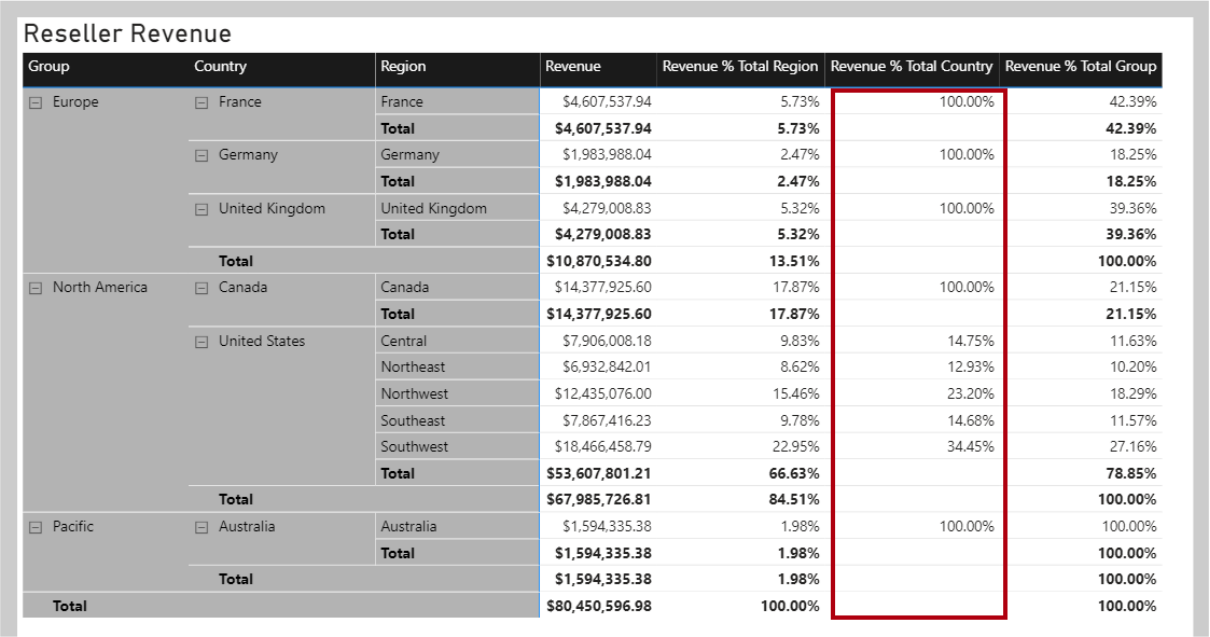
To produce a total, you will need to use an iterator function, which is explained later in this module.

Three other functions that you can use to test filter state are:

- ISFILTERED - Returns TRUE when a passed-in column reference is *directly* filtered.
- ISCROSSFILTERED - Returns TRUE when a passed-in column reference is *indirectly* filtered. A column is cross-filtered when a filter that is applied to another column in the same table, or in a related table, affects the reference column by filtering it.
- ISINSCOPE - Returns TRUE when a passed-in column reference is the level in a hierarchy of levels.

6. Return to **Page 2** of the report, and then modify the **Revenue % Total Country** measure definition to test that the **Region** column in the **Sales Territory** table is in scope. If it's not in scope, the measure result should be BLANK.

```
Revenue % Total Country =
VAR CurrentRegionRevenue = [Revenue]
VAR TotalCountryRevenue =
    CALCULATE(
        [Revenue],
        REMOVEFILTERS('Sales Territory'[Region])
    )
RETURN
    IF(
        ISINSCOPE('Sales Territory'[Region]),
        DIVIDE(
            CurrentRegionRevenue,
            TotalCountryRevenue
        )
    )
```



Reseller Revenue

Group	Country	Region	Revenue	Revenue % Total Region	Revenue % Total Country	Revenue % Total Group
Europe	France	France	$4,607,537.94	5.73%	100.00%	42.39%
		Total	**$4,607,537.94**	**5.73%**		**42.39%**
	Germany	Germany	$1,983,988.04	2.47%	100.00%	18.25%
		Total	**$1,983,988.04**	**2.47%**		**18.25%**
	United Kingdom	United Kingdom	$4,279,008.83	5.32%	100.00%	39.36%
		Total	**$4,279,008.83**	**5.32%**		**39.36%**
	Total		**$10,870,534.80**	**13.51%**		**100.00%**
North America	Canada	Canada	$14,377,925.60	17.87%	100.00%	21.15%
		Total	**$14,377,925.60**	**17.87%**		**21.15%**
	United States	Central	$7,906,008.18	9.83%	14.75%	11.63%
		Northeast	$6,932,842.01	8.62%	12.93%	10.20%
		Northwest	$12,435,076.00	15.46%	23.20%	18.29%
		Southeast	$7,867,416.23	9.78%	14.68%	11.57%
		Southwest	$18,466,458.79	22.95%	34.45%	27.16%
		Total	**$53,607,801.21**	**66.63%**		**78.85%**
	Total		**$67,985,726.81**	**84.51%**		**100.00%**
Pacific	Australia	Australia	$1,594,335.38	1.98%	100.00%	100.00%
		Total	**$1,594,335.38**	**1.98%**		**100.00%**
	Total		**$1,594,335.38**	**1.98%**		**100.00%**
Total			**$80,450,596.98**	**100.00%**		**100.00%**

In the matrix visual, notice that **Revenue % Total Country** values are now only displayed when a region is in scope.

Perform context transition

What happens when a measure or measure expression is evaluated within row context? This scenario can happen in a calculated column formula or when an expression in an iterator function is evaluated.

Exercise 09: Context Transaction

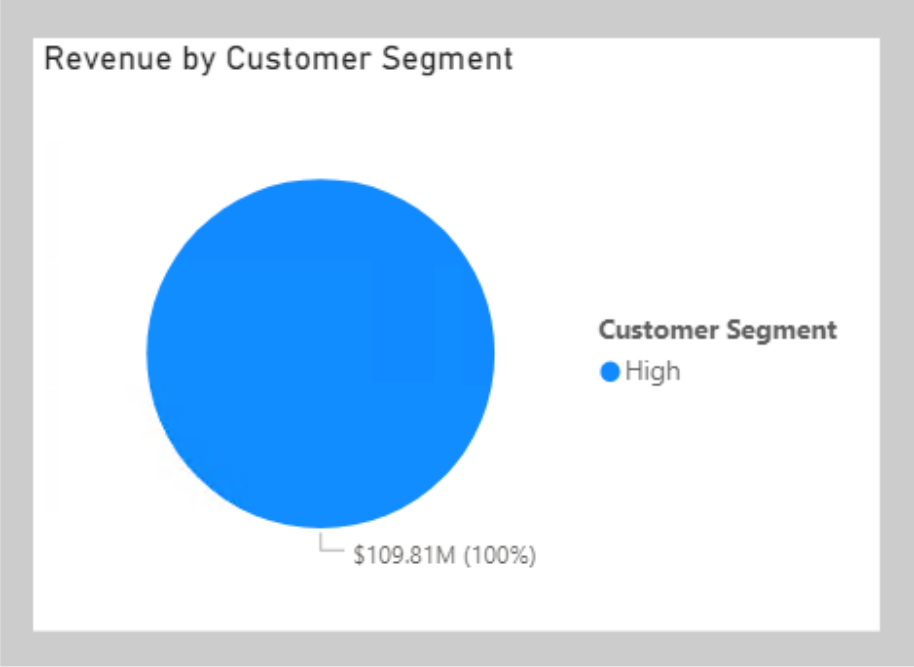
Use file 02 Start.pbix

In this exercise you will add a calculated column to the **Customer** table to classify customers into a loyalty class.

The scenario is simple: When the revenue that is produced by the customer is less than $2500, the customer is classified as **Low**; otherwise they're classified as **High**.

```
Customer Segment =
VAR CustomerRevenue = SUM(Sales[Sales Amount])
RETURN
    IF(CustomerRevenue < 2500, "Low", "High")
```

On **Page 4** of the report, add the **Customer Segment** column as the legend of the pie chart.



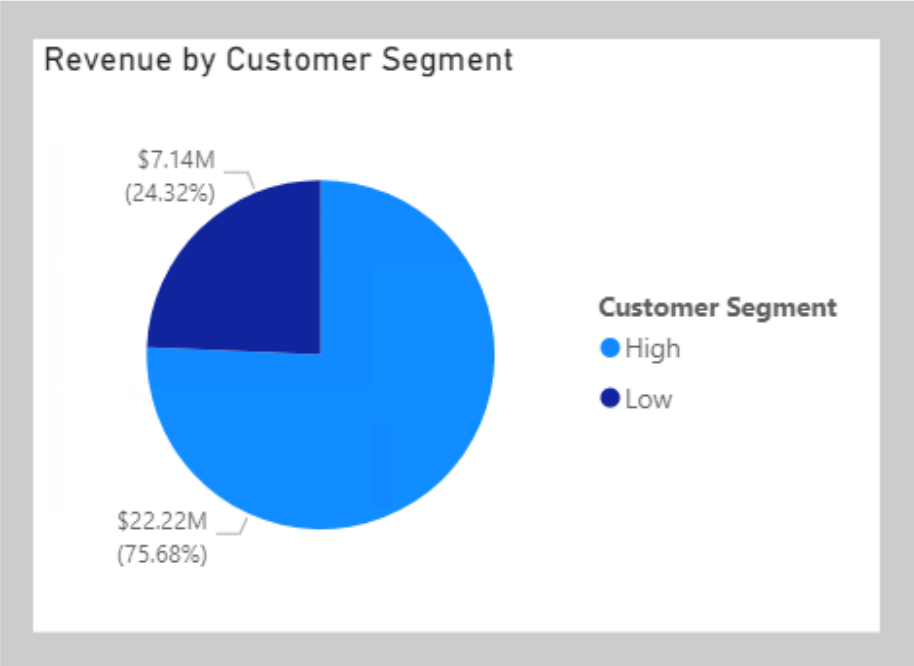
Notice that only one **Customer Segment** value exists. The reason is because the calculated column formula produces an incorrect result: Each customer is assigned the value of **High** because the expression `SUM(Sales[Sales Amount])` isn't evaluated in a filter context. Consequently, each customer is assessed on the sum of *every* **Sales Amount** column value in the **Sales** table.

To force the evaluation of the `SUM(Sales[Sales Amount])` expression *for each customer*, a **context transition** must take place that applies the row context column values to filter context. You can accomplish this transition by using the `CALCULATE` function without passing in filter expressions.

Modify the calculated column definition so that it produces the correct result.

```
Customer Segment =
VAR CustomerRevenue = CALCULATE(SUM(Sales[Sales Amount]))
RETURN
    IF(CustomerRevenue < 2500, "Low", "High")
```

In the pie chart visual add the new calculated column to the Legend well, verify that two pie segments now display.



In this case, the `CALCULATE` function applies row context values as filters, known as ***context transition***.

To be accurate, the process doesn't quite work that way when a unique column is on the table. When a unique column is on the table, you only need to apply a filter on that column to make the transition happen. In this case, Power BI applies a filter on the **CustomerKey** column for the value in row context.

If you reference measures in an expression that's evaluated in row context, context transition is automatic. Thus, you don't need to pass measure references to the `CALCULATE` function.

Modify the calculated column definition, which references the **Revenue** measure, and notice that it continues to produce the correct result.

```
Customer Segment =
VAR CustomerRevenue = [Revenue]
RETURN
    IF(CustomerRevenue < 2500, "Low", "High")
```

Now, you can complete the **Sales Commission** measure formula. To produce a total, you need to use an iterator function to iterate over all regions in filter context. The iterator function expression must use the `CALCULATE` function to transition the row

context to the filter context. Notice that it no longer needs to test whether a single **Country** column value in the **Sales Territory** table is in filter context because it's known to be filtering by a single country/region (because it's iterating over the regions in filter context and a region belongs to only one country/region).

Switch to **Page 6** of the report, and then modify the **Sales Commission** measure definition to use the `SUMX` iterator function:

```
Sales Commission =
SUMX(
    VALUES('Sales Territory'[Region]),
    CALCULATE(
        [Revenue]
        * IF(
            VALUES('Sales Territory'[Country]) = "United States",
            0.15,
            0.1
        )
    )
)
```

The table visual now displays a sales commission total for all regions.

Region	Revenue	Sales Commission
Australia	$10,655,335.96	$1,065,533.60
Canada	$16,355,770.46	$1,635,577.05
Central	$7,909,009.01	$1,186,351.35
France	$7,251,555.65	$725,155.56
Germany	$4,878,300.38	$487,830.04
Northeast	$6,939,374.48	$1,040,906.17
Northwest	$16,084,942.55	$2,412,741.38
Southeast	$7,879,655.07	$1,181,948.26
Southwest	$24,184,609.60	$3,627,691.44
United Kingdom	$7,670,721.04	$767,072.10
Total	**$109,809,274.20**	**$14,130,806.96**